

OBJECT ORIENTED PROGRAMMING USING JAVA**Time : 2:30 Hours]****[Maximum Marks : 50****[Minimum Marks : 17****NOTES:**

- i) Attempt all questions.
- ii) Students are advised to specially check the Numerical Data of question paper in both versions. If there is any difference in Hindi Translation of any question, the students should answer the question according to the English version.
- iii) Use of Pager and Mobile Phone by the students is not allowed.

Q1) Answer any two of the following.**[2 × 5 = 10]**

- a) Explain the various features of Java.
- b) Write a difference between abstract class and interface.
- c) What are the different control flow statements available in Java?

Q2) Answer any two of the following.**[2 × 5 = 10]**

- a) What is an Exception Handling? How it is handled in Java?
- b) How packages are created and imported in Java?
- c) What is inheritance? What are its types?

Q3) Answer any two of the following.**[2 × 5 = 10]**

- a) What is constructor? Write its various properties.
- b) Write a Java program to find whether a number is odd or even.
- c) Differentiate between upcasting and downcasting.

Q4) Answer any two of the following.**[2 × 5 = 10]**

- a) Explain the life cycle of a thread.
- b) Write a short note on Datatypes in Java.
- c) Write a Java program to find factorial of a given number.

Q5) Write a short note on any two of the following.**[2 × 5 = 10]**

- a) Polymorphism
- b) Access specifies / modifiers
- c) Thread priorities

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Q1) Answer any two of the following.**[2 × 5 = 10]**

- a) Explain inheritance and encapsulation in Java.
- b) Differentiate between Procedure Oriented Programming and Objected Oriented Programming.
- c) Explain structure of Java program.

Q2) Answer any two of the following.**[2 × 5 = 10]**

- a) Explain operators in Java.
- b) Explain switch case in Java with examples.
- c) Explain data types in Java in detail.

Q3) Answer any two of the following.**[2 × 5 = 10]**

- a) Explain parameterised constructor with example.
- b) Write short notes on package and string.
- c) Write a program using java to print even number.

Q4) Answer any two of the following.**[2 × 5 = 10]**

- a) Define access control in detail.
- b) Explain abstract class with example.
- c) Differentiate between classes and interface.

Q5) Answer any two of the following.**[2 × 5 = 10]**

- a) Explain multi-threading in detail.
- b) Explain method overloading.
- c) Explain Try-Catch-Finally Block with suitable example.

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Q1) Attempt any two of the following. [2×5=10]

- a) Write a difference between object oriented and procedure oriented programming.
- b) Explain different data types used in java.
- c) Write and explain the syntax of creating an interface in Java.

Q2) Attempt any two of the following. [2×5=10]

- a) Write a program in Java to implement 'switch-case' statement.
- b) Write a short note on:
 - i) Data abstraction
 - ii) Data encapsulation
- c) Java is a platform independent language. Justify.

Q3) Attempt any two of the following: [2×5=10]

- a) Write a difference between static and non-static variables.
- b) Explain different types of inheritance.
- c) What is constructor overloading? Explain with example.

Q4) Attempt any two of the following. [2×5=10]

- a) Discuss the importance of exception handling.
- b) What do you mean by threads? How can you create threads in Java.
- c) Explain the implementation of multiple inheritance through interface.

Q5) Attempt any two of the following: [2×5=10]

- a) Write a program in Java to calculate the area of a circle.
- b) Write a difference between multi threading & multi tasking.
- c) How would you create and access a package in Java?

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Q1) Answer any two of the following

[2×5=10]

- a) What do you mean by OOP? Explain OOP concepts.
- b) What are the different types of operators used in Java?
- c) Explain switch statement with example.

Q2) Answer any two of the following :

[2×5=10]

- a) What are constructors? Explain different types of constructors in Java.
- b) Explain about private, public and protected access control.
- c) How to import a package? Explain with example.

Q3) Answer any two of the following:

[2×5=10]

- a) What is an interface? List the rules to create an interface in Java with example.
- b) What is polymorphism? Discuss its types.
- c) What is multiple inheritance and how to perform it in Java?

Q4) Answer any two of following:

[2×5=10]

- a) List Java's Built-in Exception. Write the importance of finally block.
- b) What is exception? Explain the types of exception handling techniques in java.
- c) What is the difference between multitasking and multithreading?

Q5) Write short notes on any two of the following:

[2×5=10]

- a) Thread priority.
- b) Encapsulation.
- c) Abstract class.

Code No. : 2208

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Q1) Answer any two of the following :

[2×5=10]

- a) What are classes and object. Explain with example.
- b) Write notes on Abstraction and Encapsulation.
- c) Explain input using scanner class.

Q2) Answer any two of the following :

[2×5=10]

- a) Write a Java program to find the prime number.
- b) Write notes on string and string Buffer.
- c) Explain static and nonstatic variable methods defining packages.

Q3) Answer any two of the following :

[2×5=10]

- a) Define Inheritance. Explain single and multilevel Inheritance with example.
- b) What is Hybrid inheritance and hierarchical inheritance.
- c) What is order of invocation.

Q4) Answer any two of the following :

[2×5=10]

- a) What is Abstract class and interface.
- b) What is difference method overriding and overloading.
- c) What is upcasting and downcasting.

Q5) Answer any two of the following :

[2×5=10]

- a) How creat own exception? Explain.
- b) What is the importance of exception Handling?
- c) Write notes on : try, catches and finally.